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Unit 4 : Session 2 : SLO 2 | Built-in Higher-Order Functions (map, reduce, filter)

I. Fill in the Blanks

1. The filter() function in Python returns an object that contains elements from the iterable that satisfy a condition (predicate function).
2. A function is pure if it does not have side effects and does not change any external (global) state.

II. Multiple Choice

1. list(map(lambda x: x + 1, [1, 2, 3])) → [2, 3, 4] → map applies the lambda to each element, adding 1 to each.

III. Extra Examples

- filter: list(filter(lambda x: x % 2 == 0, range(10))) → [0, 2, 4, 6, 8].
- reduce: functools.reduce(lambda a, b: a * b, [1, 2, 3, 4]) → 24 (factorial of 4).